

HAQ Lab
Hannigan Air Quality Lab



Lesson 2: How Do We Understand Soil Quality?

Central SQ Slide Deck

Last Updated September 8, 2026 by Percy Smith

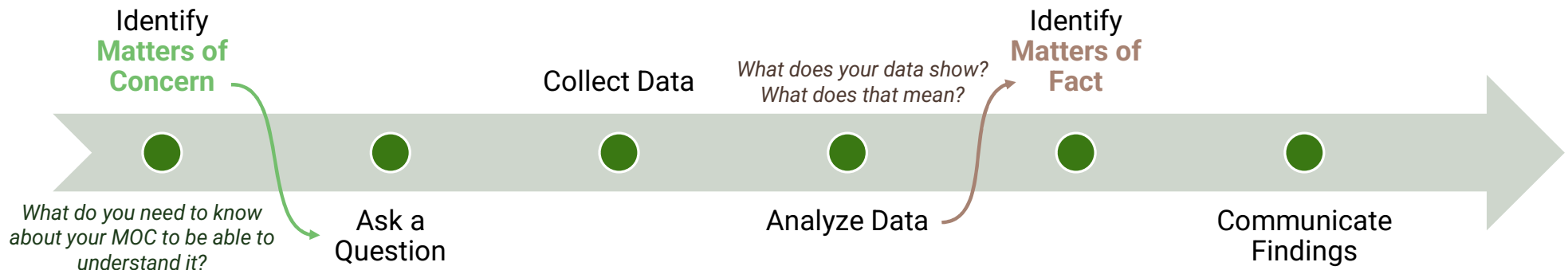


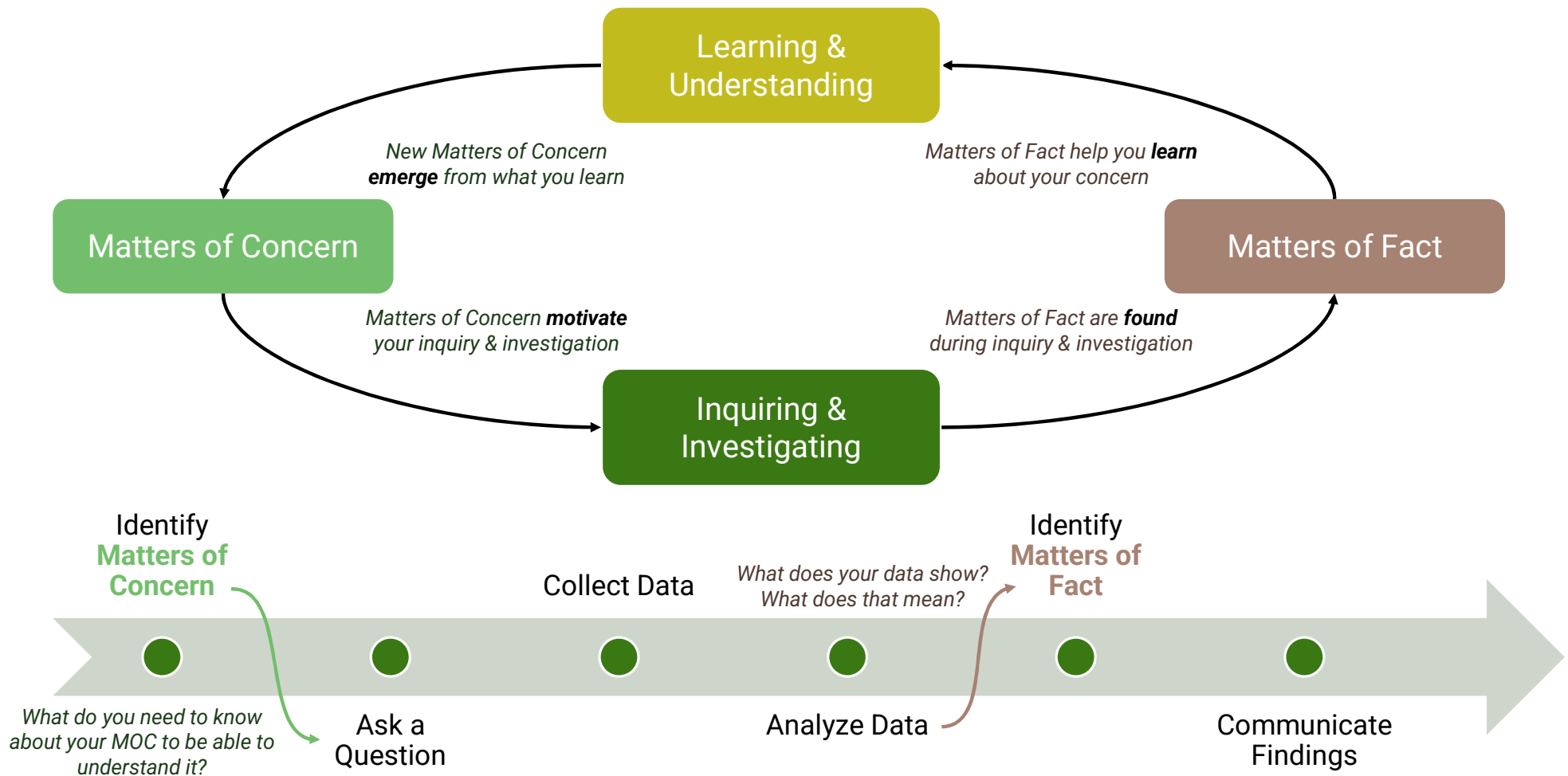
Engineering & Applied Science
UNIVERSITY OF COLORADO **BOULDER**



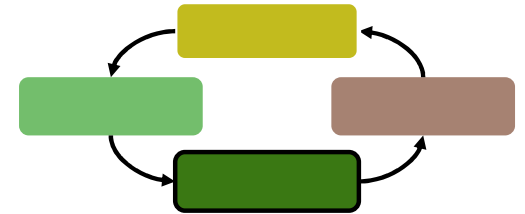
Soil Quality InQuiry: MOC & MOF

- **Matters of Concern:** Issues related to a topic that you want to understand
 - Soil quality issues that are important to you
- **Matters of Fact:** Conclusions, information & experimental results that define an issue
 - “Knowns” that you use to understand something





Qualitative & Quantitative Data



Quantitative Data

Quantify: to measure or count

Aims to answer “what” & “how much” with numerical data

Qualitative Data

Qualify: to note qualities or features

Aims to explore “how” & “why” through descriptive data



Integrating Both!

Quantitative Data

Quantify: to measure or count
Aims to answer “what” & “how much”
with numerical data

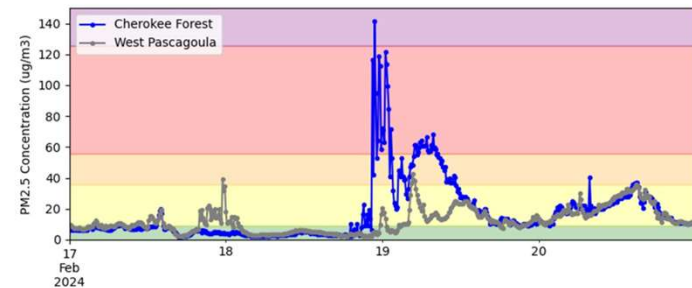


Figure 6. PM2.5 concentrations ($\mu\text{g m}^{-3}$) in the Cherokee Forest subdivision and in West Pascagoula. Elevated PM2.5 concentrations in Cherokee Forest on the evening of February 18th through the 19th correspond to symptom complaints reported by resident participants. Color background on the figure corresponds to the EPA air quality index (AQI) for PM2.5.

Qualitative Data

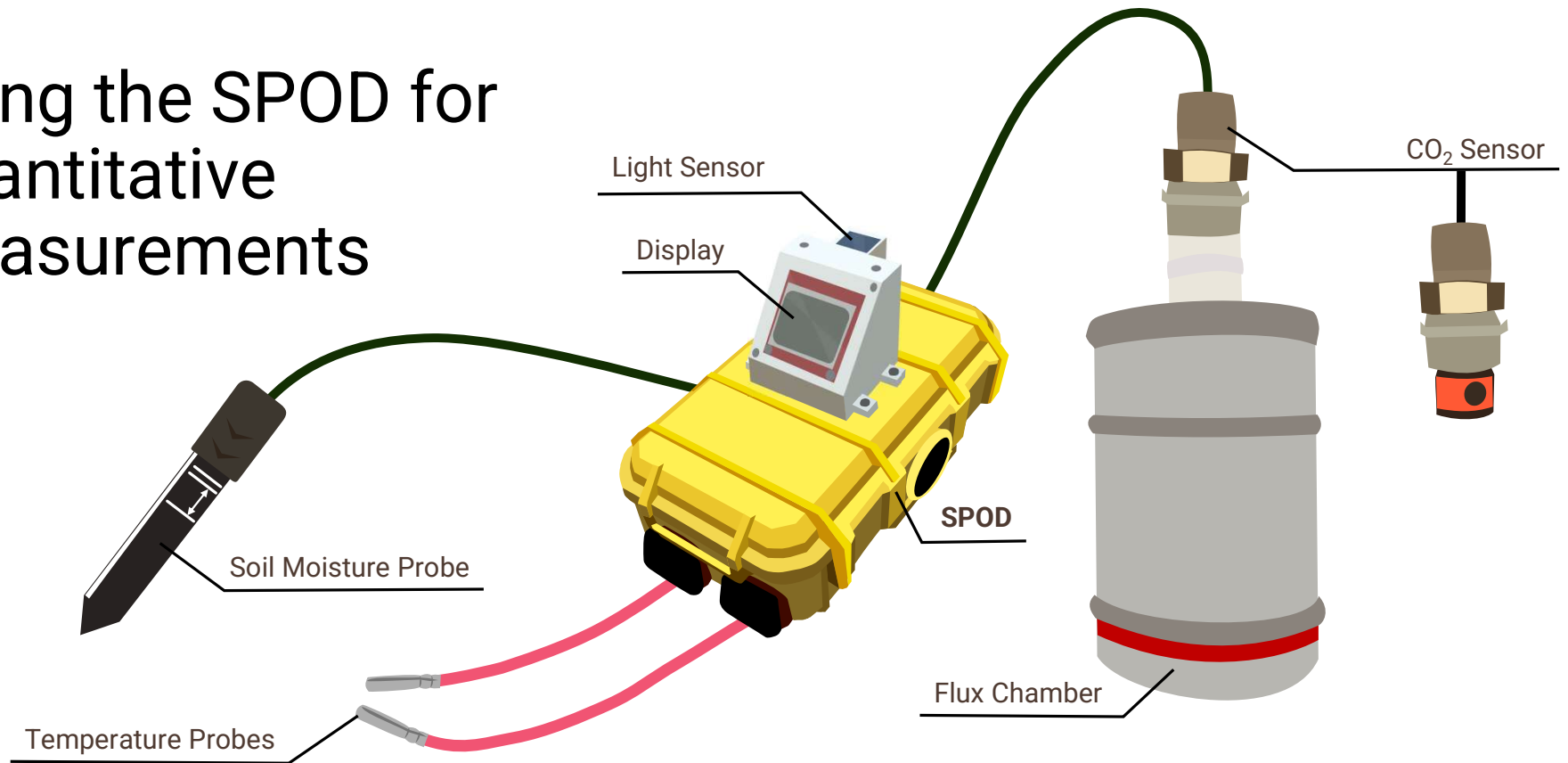
Qualify: to note qualities or features
Aims to explore “how” & “why” through
descriptive data

Date and time range of the complaint	Who noticed the smell or felt the symptom?	Where was the reporting person and what were they doing?	What is the smell like (if applicable)?	Smell intensity from 1 (very mild) to 5 (very intense)	What symptoms are felt (if applicable)?	Symptom intensity from 1 (very mild) to 5 (very intense)
2-8-24 7 PM	Myself ☒ Spouse/partner ☐ Child/grandchild ☐ Other:	Truck garage outside.	☒ Rotten eggs ☐ Cat urine/ammonia ☐ Garbage ☐ Acid ☐ Plastic burning ☐ Gasoline or fuel ☐ Sweet smell ☐ Other:	5	☐ Headache ☒ Itchy/burning eyes ☐ Skin itching/rash ☐ Dizziness/lightheadedness ☐ Sinus pain ☒ Sore throat ☐ Fatigue ☐ Difficulty breathing ☐ Other:	4
Anything else you would like to share: Strong Smell. Chemical smell.						

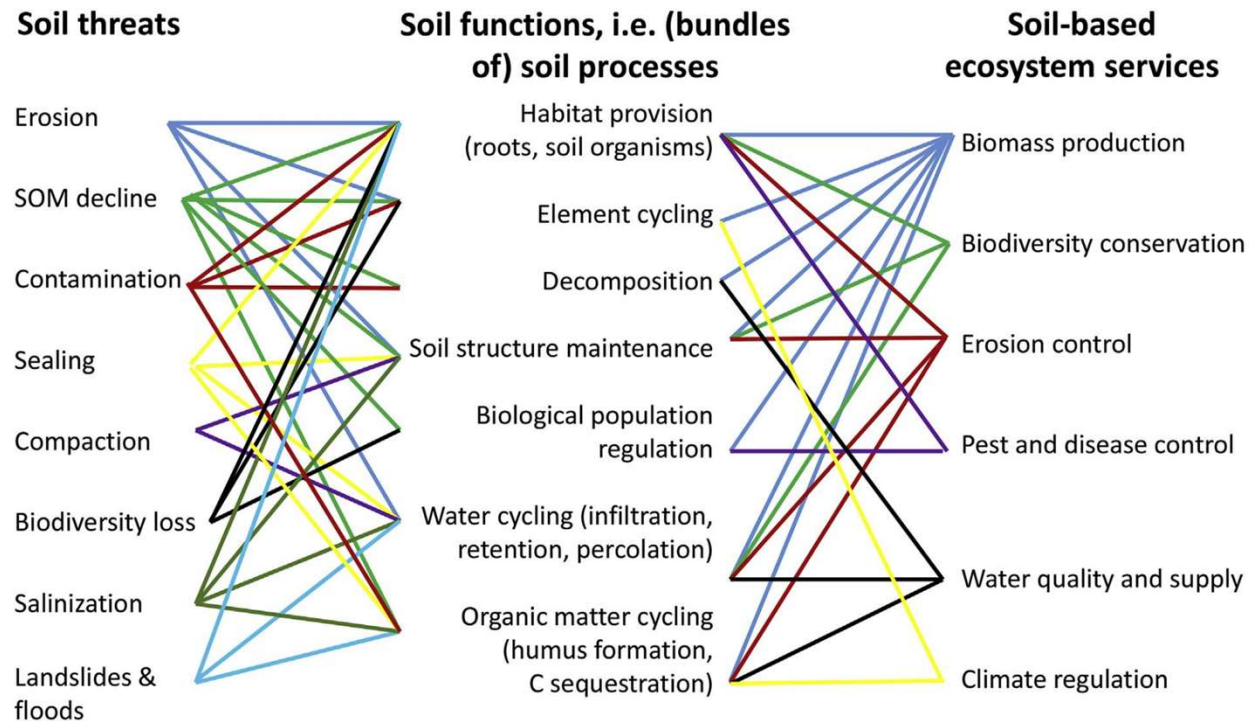
Figure 3. An example symptom and odor log entry.



Using the SPOD for Quantitative Measurements



Numerous Soil Quality Concerns...



Soil Erosion

- **Geological Erosion:** The wearing away of the land surface by natural forces
 - Process is slower than the rate of soil formation
- **Accelerated Erosion:** The wearing away of land driven by human activities
 - Exceeds the rate of soil formation



Figure 5. Example of accelerated erosion due to excess of tourist trampling in a beech (*Fagus sylvatica* L.) forest near the Saint Francis's Sanctuary of La Verna, Chiusi della Verna Municipality, Arezzo, Italy. In an area of about 1000 m² at about 1200 m above sea level, about half meter of soil has been removed.



Impacts of Accelerated Erosion

- Degradation of agricultural land
 - Decreased crop yield
- Particulate emissions
- Water contamination
- Decreased stability of land
 - Damaged infrastructure
 - Instability during floods



Figure 6.3. Drought and poor soil health created wind and water erosion during the Dust Bowl. Photo by USDA.



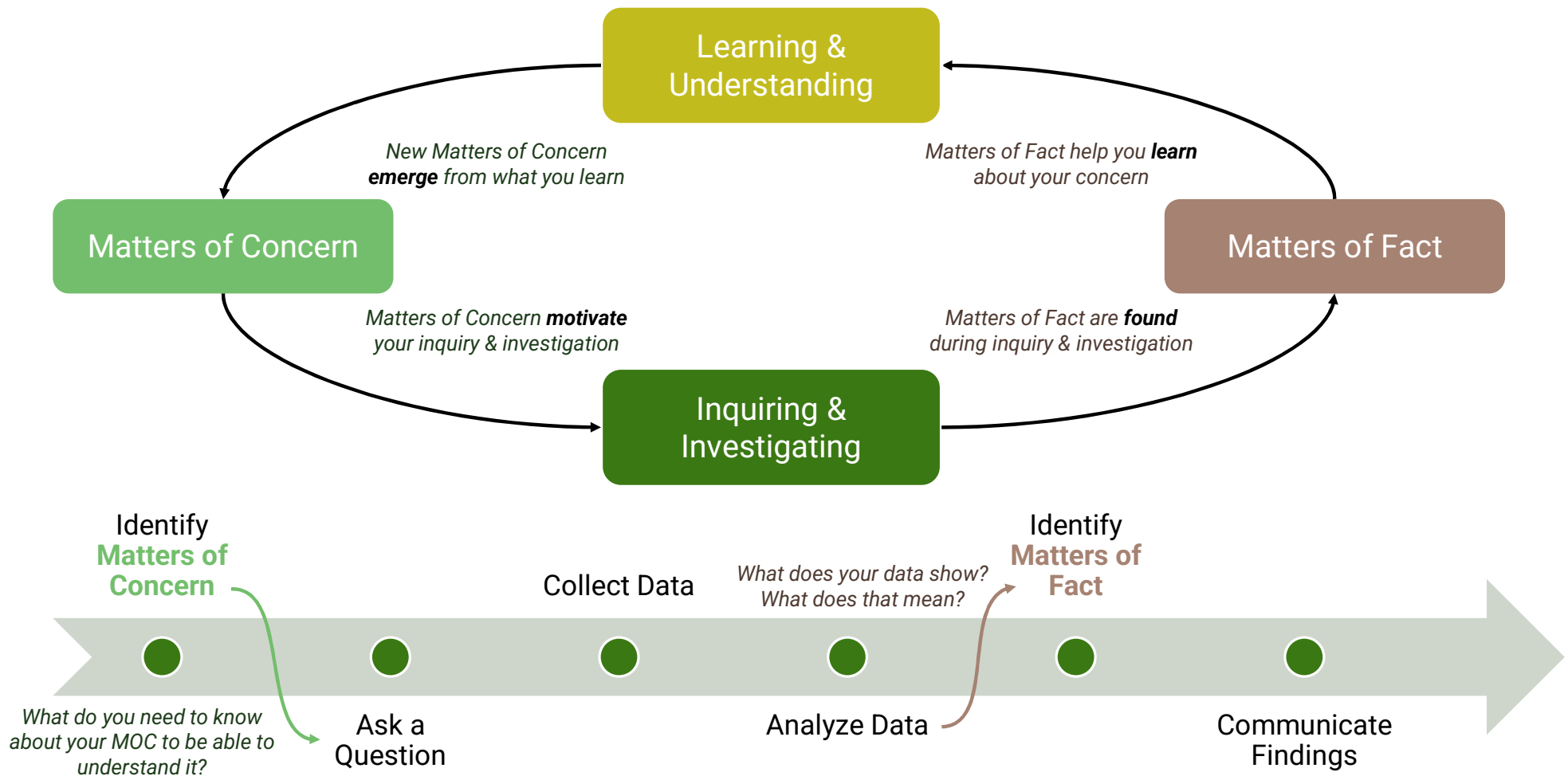
Managing Soil Erosion

- **Structural Measures:** Creation of structures that reduce erosion, often engineering structures to help stabilize the land
 - In conservation, used to complement agronomic practices
- **Agronomic Practices:** Changes to soil and crop management that build soil stability over time



Figure 1. Example of forest land set-up setting: stabilized natural ditches to slow down the deepening of the gulch. Casentino Forests, Arezzo, Italy.





If you want to learn more, we recommend starting with these excellent sources!

*Chiartas, J., Ward, A., Zahangir, K., Santih, S., Massenzio, E., Lew, S., Gomes, A., & Le, B. (2015). *Soil life*. UC Davis. Soil Life. <https://www.soillife.org/>*

Cocco, S., Cardelli, V., Corti, G., Serrani, D., Rafael, R., Dazzi, C., & Lo Papa, G. (2020). Role of land set-up systems on soil (physicochemical) conditions. *Italian Journal of Agronomy*, 15, 1766. <https://doi.org/10.4081/ija.2020.1766>

Magdoff, F., & Van Es, H. (2021). *Building Soils for Better Crops: Ecological Management for Healthy Soils* (Fourth). <https://doi.org/10.1097/00010694-199311000-00014>

Moebius-Clune, B. N. (2016). *Comprehensive assessment of soil health: The Cornell framework manual* (Third edition). Cornell University. <https://soilhealth.cals.cornell.edu/manual/>

Walkinshaw, M., O'Green, A. T., & Beaudette, D. E. (2025). *Soil Properties* [Map]. California Soil Resources Lab. <https://casoilresource.lawr.ucdavis.edu/soil-properties/>

